<table>
<tr><td>1(a)(i)</td><td>(halogens can) remove electron(s) (from a reactant)</td><td>1</td></tr>
<tr><td>1(a)(iii)</td><td>purple (gas) disappears</td><td>1</td></tr>
<tr><td>1(b)(i)</td><td>M1 Br—Br is weaker (than Cl—Cl) AND Br atoms are larger

M2 less (significant) overlap of orbitals (in Br-Br)</td><td>2</td></tr>
<tr><td>1(b)(ii)</td><td>HCl is more thermally stable</td><td>1</td></tr>
<tr><td>1(c)(i)</td><td>
<table>
<tr><th></th><th>NaCl</th><th>NaBr</th><th>NaI</th></tr>
<tr><td>with $Ag^+(aq)$</td><td>white precipitate</td><td>M1 cream / off-white ppt</td><td>M2 yellow ppt</td></tr>
<tr><td>with conc. H_2SO_4</td><td>acid – base</td><td>acid – base,
then redox</td><td>acid – base,
then redox</td></tr>
<tr><td>observations with conc. H_2SO_4</td><td>M3 steamy fumes
OR
fizzing</td><td>M4 brown liquid
OR
fizzing</td><td>• black solid
• yellow solid
• effervescence</td></tr>
</table>
</td><td>4</td></tr>
<tr><td>1(c)(ii)</td><td>M1 black solid = iodine / I_2 AND yellow solid = sulfur / $S_{(8)}$

M2 effervescence = hydrogen sulfide / $H_2S(g)$</td><td>2</td></tr>
</table>

Examiner report missing
9701_w25_qp_24 Question 1

3(a)(i)	0.00658(0) mol	**1**
3(c)	2.46	**1**
3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	**1**
3(d)(ii)	**M1** HCl is more stable **AND** (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length **M2** (In HCl) stronger attraction between nucleus and shared pair of electrons **OR** (In HCl) (fewer shells of electrons so) less shielding **OR** (In HCl) greater orbital overlap so stronger interaction	**2**
3(d)(iii)	any value or range of values for K_c between $5.50 \times 10^{-34} < K_c < 1.26 \times 10^{-3}$	**1**

Examiner report missing
9701_s25_qp_24 Question 3

6(a)(i)	two or more reactants / molecules react to make (only) one product	**1**
6(a)(ii)	mix / add Cl_2 into water	**1**
6(a)(iii)	water purification / killing bacteria or microbes	**1**

Examiner report missing
9701_s25_qp_23 Question 6

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	**1**
3(a)(ii)	• atoms larger (down the group) • distance between electron pair and nucleus greater / longer bond length / lesser orbital overlap • weaker attraction	**2**
3(b)	$Cl_2 + H_2O \rightarrow HCl + HOCl$ ClO^- / $HOCl$ kills bacteria	**2**
3(c)(i)	sodium iodide / NaI	**1**
3(c)(ii)	sodium chloride / NaCl	**1**
3(c)(iii)	cream(-coloured) / off-white precipitate / solid forms	**1**

Examiner report missing
9701_m25_qp_22 Question 3

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(ii)	oxidising strength **decreases** from chlorine to iodine	**1**
2(a)(iii)	green colour disappears	**1**
2(b)(ii)	ultraviolet	**1**
2(b)(iii)	$C_2H_6 + Cl^{\bullet} \rightarrow C_2H_5^{\bullet} + HCl$	**1**
	$C_2H_5^{\bullet} + Cl_2 \rightarrow C_2H_5Cl + Cl^{\bullet}$	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(ii)	oxidising strength **decreases** from chlorine to iodine	**1**
2(a)(iii)	green colour disappears	**1**
2(a)(iv)	H—Hal covalent bond strength **decreases** down the group	**1**
2(b)(ii)	ultraviolet	**1**
2(c)(i)	**cold** NaOH(aq)	**1**

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1(a)	<table><tr><td>halogen</td><td>colour at 293 K</td></tr><tr><td>chlorine</td><td>green</td></tr><tr><td>bromine</td><td>brown</td></tr><tr><td>iodine</td><td>(dark) grey</td></tr></table> All three correct for one mark	**1**
1(b)	**M1** (it / volatility) decreases **M2** more electrons in molecules **M3** (so) increased strength of instantaneous dipole – induced dipole forces (of attraction)	**3**
1(c)(i)	$Br_2 + 2I^- \rightarrow 2Br^- + I_2$	**1**
1(c)(ii)	(Br_2 / bromine is an) oxidising agent **AND** removes electron(s) from iodide (ions) I^- **OR** increases the oxidation number of I^- / iodide (ion)	**1**
1(d)(i)	$NaBr + H_2SO_4 \rightarrow HBr + NaHSO_4$ **OR** $2NaBr + H_2SO_4 \rightarrow 2HBr + Na_2SO_4$	**1**
1(d)(ii)	**M1** sodium chloride **M2** Cl^- is not a strong enough reducing agent to reduce H_2SO_4 **M3** Br^- & I^- are strong enough reducing agents to react (further) with (the S in) H_2SO_4 *Alternative M2 and M3* **M2** ability to behave as reducing agents increases down the group **M3 EITHER** Cl^- is not strong enough to reduce H_2SO_4 **OR** Br^- and I^- react (further) with (the S in) H_2SO_4	**3**

Question 1

(a) The colour of the halogens (at 293 K) was not often answered correctly by candidates. Common errors included chlorine described as yellow and iodine as a purple gas. Several candidates gave the colours of silver halides.

(b) The volatility of the halogens chlorine, bromine and iodine is associated specifically with the increased strength of the instantaneous dipole-induced dipole forces between the molecules due to the increase in the number of electrons in these molecules. Many candidates correctly stated that the volatility decreases from chlorine to bromine to iodine. This trend was then incorrectly attributed or linked to the electronegativity variations of the elements or the attraction between the nucleus and the outer electrons.

(c) (i) The majority of candidates used the full formula equation rather than ionic for the reaction of bromine with sodium iodide. Many answers included the spectator ion Na^+ in the equation.

(ii) It was recognised by several candidates that bromine, when reacted with sodium iodide, behaved as an oxidising agent to form iodine and a bromide. Many explanations then incorrectly referred to a change in oxidation number, or movement of electrons, in iodine instead of the iodide ions.

(d) (i) Many candidates gave a correct equation for the reaction of NaBr with concentrated sulfuric acid. Common errors were $NaSO_4$, for sodium sulfate(VI).

(ii) Candidates often identified NaCl as the salt that produced the largest percentage of hydrogen halide. The explanations frequently referred to the reactivity of the sodium halides instead of describing the trend in the hydrogen halides or halide ions as reducing agents.

The most common incorrect answers identified ‘sodium iodide’ producing the largest percentage yield of hydrogen halide as ‘it is the most reactive’.

2(a)	<table><tr><td>formula of chloride</td><td>● $NaCl$</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**
2(b)(i)	disproportionation	**1**
2(b)(ii)	$NaClO$ / sodium chlorate(I)	**1**
2(b)(iii)	$3Cl_2 + 6NaOH \rightarrow NaClO_3 + 5NaCl + 3H_2O$	**1**
2(b)(iv)	white ppt / solid	**1**
	dissolves (in $NH_3(aq)$)	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(c)	(Br_2) no visible reaction OR colourless to orange / brown / yellow solution	**1**
	(conc. H_2SO_4) no visible reaction OR Solution remains colourless	**1**
	(dil. $AgNO_3$) white precipitate forms	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

Most answers stated that the observation from the addition of concentrated H_2SO_4 to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

(d) (i) This question provided information that the chemical behaviour of $POCl_3$ is similar to that of PCl_5; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

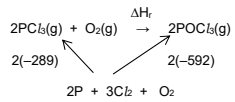
(e) (i) Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:



ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

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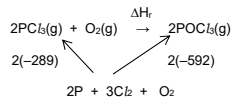
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496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

4(a)(i)	$2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$ **OR** $KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$	**1**

4(a)(ii)	**M1** HI(g) is not the only gaseous product **OR** SO_2 and / or H_2S and / or I_2(g) is (also) made **M2** HI / iodide (ion) is oxidised **OR** H_2SO_4 oxidises HI **OR** HI reduces sulfuric acid	**2**

4(b)(i)	(from colourless) to purple	**1**

Question 4

(a) (i) Some good answers were seen.

(ii) Very few candidates knew that hydrogen iodide is not prepared in the same way as hydrogen chloride because the hydrogen iodide and concentrated H_2SO_4 would undergo a redox reaction to form a variety of products to include SO_2, H_2S and I_2.

(b) (i) A small number of candidates recognised that when a sample of colourless HI(g) is placed in a sealed container and allowed to reach equilibrium the colour changes to purple, attributed to the formation of I_2(g).

(ii) This question was well answered by many candidates.

(iii) The calculation of the concentration of iodine in an equilibrium mixture was generally not well answered. Candidates who deduced that [HI] is (1.70 ÷ 2 =) 0.85 mol in the equilibrium mixture tended to gain some credit. Very few used the expression $\sqrt{(0.0217 \times 0.85)}$ to calculate the $[I_2]$ as 0.125 mol dm^{-3}.

(c) (i) Some candidates were able to describe the difference in composition of the equilibrium mixture at the two temperatures.

(ii) Some candidates deduced that the reaction is endothermic in the forward direction leading to a lower concentration of H_2 and I_2 in the equilibrium mixture.

2(a)(i)	<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>All correct for one mark	**1**
2(d)(i)	$Cl_2 + H_2O \rightleftharpoons HOCl + HCl$	**1**
2(d)(ii)	$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$	**1**
2(e)(i)	ClO^-	**1**

Question 2

(a) (i) The maximum oxidation numbers of the Period 3 chlorides were well known.

(ii) The majority of responses were correct. Some candidates produced responses that were vague and incomplete. Occasionally, explanations were incorrectly given in terms of changes to the nuclear charge.

(b) (i) The majority of candidates identified the correct formula of the silicon chloride and gave an appropriate balanced equation for the reaction.

(ii) A significant proportion of candidates incorrectly described the pH of the solution as greater than or equal to 7.

(c) (i) The equation to represent the reaction for phosphorus(V) chloride with water was less well known than the equation for **(b)(ii)**. The production of phosphorus oxide rather than phosphoric acid was a common incorrect answer. Other equations which showed appropriate species were not correctly balanced.

(ii) Many candidates gave appropriate suggestions for the pH of the solution produced in **(c)(i)**.

(d) (i) Equations using C*l* rather than C*l*$_2$ and production of O_2 with HC*l* were common incorrect answers.

(ii) A small proportion of candidates identified all the correct species and produced a balanced equation. Some candidates found the equation for the reaction of chlorine with hot NaOH(aq) to be demanding.

(e) (i) A wide range of incorrect species were seen in the answers; many of which were not ions.

(ii) Many correct equations representing this reaction were seen. Occasionally, equations gave incorrect multiples for the correct species which included the decomposition of some hydrogen peroxide without the use of sodium chlorate(I).

(iii) A significant proportion of candidates used the appropriate data to calculate the number of moles of sodium chlorate(I) used in a single titration. Difficulties arose in attempts to work out the concentration in the original sample, **W**. Occasionally, concentrations were left in mol dm^{-3} rather than converting this into g dm^{-3}, as detailed in the question.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**
3(c)(i)	hot aqueous sodium hydroxide / hot NaOH(aq)	**1**
3(c)(ii)	oxidation number / state of chlorine / C*l* increases / goes (from 0) to +5 **AND** decreases / goes (from 0) to –1	**1**

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

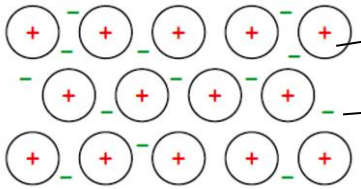
(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

2(a)(i)	 **M1** diagram showing minimum of 4 particles (in total in 2 rows) • circles containing X^{n+} do not have to be labelled •  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled electrons	**1**
	M2 label / legend showing **delocalised** electrons	**1**

2(d)(i)	white precipitate yellow solid (dark) grey solid / purple gas effervescence / misty fumes rotten egg smell	**2**

2(d)(ii)	moles of $AgNO_3 = \frac{33.65}{1000} \times 0.0500$ (= $1.68(25) \times 10^{-3}$)	**1**
	moles of **X** = ½ × moles of $AgNO_3$ = 0.250 ÷ M_r(**X**) ∴ M_r(**X**) = 297.2	**1**
	A_r of Group 2 element is 297.2 – 2(79.9) = 137.4 AND **X** is $BaBr_2$	**1**

Question 2

This was an inorganic-chemistry question based around Group 2.

(a) (i) A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

(ii) There were many good answers here although some responses continued to use free electrons instead of delocalised.

(b) A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

(c) Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

(d) (i) This question was answered quite well by candidates.

(ii) A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the A_r of $1\times$ or $2 \times Cl$.